DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS
First Semester, SY 2024-2025

COURSE CODE	**:**	S-ITPC315LA
COURSE TITLE	**:**	Networking 2
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	One Unit
LEARNING MODALITY	**:**	Fully Onsite
PRE-REQUISITES	**:**	S-ITCS214/LA
CO-REQUISITES	**:**	S-ITPC315
PROFESSOR	**:**	Joneil M. Baculod
CONSULTATION HOURS	**:**	Saturday: 4PM – 5PM / online via MS Tea
CONTACT INFORMATON	**:**	jmbaculod@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Deals with both static routing and dynamic routing protocols. By examining multiple routing protocols, you will gain a better understanding of each of the individual routing protocols and a better perspective of routing in general. Learning the configuration of routing protocols is fairly simple. Developing an understanding of the routing concepts themselves is more difficult, yet is critical for implementing, verifying, and troubleshooting routing operations.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO 1. Create a fully connected LAN

CLO 2. Explain the principles of IPv4 and IPv6 address management.

CLO 3. Configure basic network security.

CLO 4. Explain how to use security best practices to mitigate attacks.

CLO 5. Analyze routing algorithms protocols, process routing tables and configure routers for proper orientation of an efficient network.

CLO 6. Create and implement static routes IPv4 and IPv6.

CLO 7. Create and implement dynamic routing protocols.

CLO 8. Test for common configuration issues.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
Midterm Period	
CLO1. Create a fully connected LAN.	TLO1. Explain the need for routing.
CLO2. Explain the principles of IPv4 and IPv6 address management.	TLO2. Explain the principles of IPv4 and IPv6 address management.
CLO3. Configure basic network security.	TLO3. Configure devices for secure remote management.
CLO4. Explain how to use security best practices to mitigate attacks.	TLO4. Explain how security tools and software updates mitigate network security threats. TLO5. Explain how antimalware software mitigates data loss and service disruptions.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Allotted Hours
Module 1 CLO 1 CLO 3 TLO 1 TLO 6	Week 1 Aug 19-24	• Gospel reading • Presentation on the synchronous flow and activities to do during asynchronous mode. • LMS Expectations: • Netacad Course Experience – Interface, Submission, Around the Online Module • Schoolbook Details for Submission of Assessment • Module 1: Routing Between Networks • Objective: Create a fully connected LAN • Lab Activities ○ IPv4 Addresses and Network Communication ○ Connect to a Wireless Router • Share the screen cap of your achievement in the quiz • Download Packet Tracer version 8.0.1	MS Teams NetAcad SchoolBook	3
Module 2 CLO 2 TLO 2	Week 2 Aug 26-31	• Gospel reading • Recap of previous meeting achievements • Module 2: The Internet Protocol • Objective: Explain the features of an IP address ○ Define the new key terms in this module ○ Discussion on IPv6 replacing IPv4 ○ Subnetting Networks – Classful and Classless IPv4 ○ Explain Features of IPv6 • Lab Activities: ○ Binary Game - Share the screen cap of your achievement in the game to show the highest level completed in 30 minutes. • Teacher-made Packet Tracer IPv4	MS Teams NetAcad SchoolBook	3

Module 2 CLO 2 TLO 2	Week 3 Sept 2-7	• Gospel reading • Module 2 continuation • Module 2: The Internet Protocol • Objective: Explain the features of an IP address ○ Define the new key terms in this module ○ Discussion on IPv6 replacing IPv4 ○ Student Presentation of Subnetting Skills • Lab Activities ○ Recap and demonstration of answer key. Some students may continue working on this PT. Share the screen cap of your Packet Tracer score showing your User Profile. • Teacher-made Packet Tracer IPv6	MS Teams NetAcad SchoolBook	3
Module 3 CLO 1 CLO 4 TLO 4 TLO 5 TLO 6	Week 4 Sept 9-14	• Gospel reading • Read Module 3: Cisco Switches and Routers • Objective: Compare in-band and out-of-band management access • Describe LAN switches, switch boot process, small business routes and the router boot process. • You may try the module provided videos and animations	MS Teams NetAcad SchoolBook	3
Module 4 CLO 4 TLO 4 TLO 5	Week 5 Sept 16-21	• Gospel reading • Read Module 4: The Cisco IOS Command Line • Objective: Explain how to use the Cisco IOS. • Summative Assessment 1 is now available online	MS Teams NetAcad SchoolBook	3

Module 4 CLO 1 TLO 1 TLO 2 TLO 3	Week 6 Sept 23-28	• Gospel reading • Module 4 continuation • Read Module 4: The Cisco IOS Command Line • Objective: Explain how to use the Cisco IOS. • You may try the module provided videos and animations • Formative Activity: ○ Packet Tracer – Navigate the IOS • Packet Tracer – Use Cisco IOS Show Commands	MS Teams NetAcad SchoolBook	3
Module 5 CLO 1 CLO 2 TLO 1 TLO 2 TLO 3 TLO 4	Week 7 Sept 30 - Oct 5	• Gospel reading • Read Module 5: Build a Small Network • Objective: Build a simple computer network using Cisco devices. • Formative Activity: ○ Packet Tracer – Implement Basic Connectivity • Packet Tracer - Configure Initial Router Settings	MS Teams NetAcad SchoolBook	3
	Week 8 Oct 7-12	Summative Assessment Midterm Laboratory Week	Schoolbook	3
TOTAL				24

Learning Outcomes for Final Period

Course Learning Outcomes	**Topic Learning Outcomes**
Final Period	
CLO1. Analyze routing algorithms protocols, process routing tables and configure routers for proper orientation of an efficient network.	TLO1. Explain how a router learns about remote networks when operating in a small to medium-sized business network
CLO2. Create and implement static routes IPv4 and IPv6.	TLO2. Configure static routes to enable connectivity in a small to medium-sized business network.

CLO3. Create and implement dynamic routing protocols.	TLO3. Explain the function of dynamic routing protocols.
CLO4. Test for common static and default route configuration issues.	TLO4. Describe some of the approaches used to troubleshoot networks TLO5. Test for static and default route configurations.

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Allotted Hours**
Module 6, 7 CLO 5 CLO 6 TLO 6 TLO 7	Week 10 Oct 21-26	• Gospel reading • Module 6, 7 Routing Concepts, Static Routing • Objective: Explain how routers use information in packets to make forwarding decisions • Check Your Understanding on Path Determination, Packet Forwarding, IP Routing Table and Compare Static Routing vs. Dynamic Routing • Understand the concept of Static, Default and Floating static route for IPv4 and IPv6 • Lab Activities: ○ Packet Tracer Configure IPv4 and IPv6 Static and Default Routes • Troubleshoot Static and Default Routes	MS Teams NetAcad SchoolBook	3
	Week 11 Oct 28 - Nov 2	***Independent Learning Week***		

Module 8 CLO 7 TLO 6 TLO 8	Week 12 Nov 4-9	• Gospel reading • Module 8 Dynamic Routing EIGRP • Objective: • Explain EIGRP Characteristics • Implement EIGRP for IPv4 • Explain how EIGRP operates in small to medium-sized networks • Implement EIGRP for IPv6 • Lab Activities: ○ Configure Basic EIGRP with IPv4 ○ Investigate DUAL FSM • Configure Basic EIGRP with IPv6	MS Teams NetAcad SchoolBook	3
Module 9 CLO 7 TLO 6 TLO 8	Week 13 Nov 11-16	• Gospel reading • Module 9 Dynamic Routing OSPF • Objective: • Explain how single-area OSPF Operates • Implement single-area OSPFv2 • Implement single-area OSPFv2 • Explain how multiarea operates • Implement multiarea OSPF • Formative Activity: ○ Configure OSPFv2 in a Single Area ○ Configure OSPFv3 in a Single Area ○ Configure multiarea OSPFv2 • Configure multiarea OSPFv3	MS Teams NetAcad SchoolBook	3
Module 10 CLO 6 TLO 9 TLO 10	Week 14 Nov18-23	• Gospel reading • Module 10 Troubleshoot Static and Default Routes • Objective: Troubleshoot static and default route configurations. • Lab Activities: ○ Syntax Checker Task ○ Check your Understanding on Packet Processing with Static Routes • Summative Assessment 1 is now available online	MS Teams NetAcad SchoolBook	3

Module 11 CLO 6 CLO 7 TLO 6 TLO 7 TLO 9 TLO 10	Week 15 Nov 25-30	• Gospel reading • Module 11 Troubleshoot Common Network Problems • Objective: Troubleshoot basic network connectivity issues • Lab Activities: ○ Syntax Checker Task • Check your Understanding on Physical Layer Problems, Common Internet Connectivity Issues	MS Teams NetAcad SchoolBook	3
Module 12 CLO 3 CLO 7 CL 8 TLO 3 TLO 9 TLO 10	Week 16 Dec 2-7	• Gospel reading • Module 11 Security Considerations • Objective: Explain how to use security best practices to mitigate attacks • Lab Activities: ○ Security Threats ○ Social Engineering ○ Types of Malware ○ Botnets and Zombies ○ Malware ○ Denial of Service ○ Distributed DoS ○ Security Tools ○ Antivirus Software • Antimalware Software	MS Teams NetAcad SchoolBook	
	Week 17 Dec 9-14	Summative Assessment Midterm Laboratory Week	SchoolBook	3
TOTAL				24

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	40%	Enabling Assessments	40%

Summative Assessment (Midterm Examination)	60%	Summative Assessment (Final Examination)	60%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
   a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
   a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES:

On-Site References

Call number or e-provider	Reference Material
TK 2102.2 .In89 2017	Cisco Networking Academy companion guide v2.1. 2017. Singapore : Pearson Education South Asia Pte Ltd
TK 2102.2487 .R762 2017	Cisco Networking Academy companion guide v6. 2017. Indianapolis, Indiana, USA : CISCO Press
TK 2101 .C737 2018	Meadows, J. 2018. Communication technology update and fundamentals. New York ; London : Routledge Taylor & Francis Group,
QA 76.73.P98 .Or82 2019	Sarker, Dr. M. 2019. Learning Python networking : a complete guide to build and deploy strong networking capabilities using Python 3.7 and Ansible. Second edition. UK : Packt Publishing, Ltd.

Online References

IEEE Xplore	Chadha, D. 2019 Software-Defined Optical Networks, IEEE. Wiley-IEEE Press
IEEE Xplore	Panek,C. 2020. Networking Fundamentals. Wiley. Wiley Data and Cybersecurity.

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1st Semester 2024-2025

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